

IoT-NDN: An IoT Architecture via Named Data Networking (NDN)

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Abstract—Internet of Things (IoT) systems have become the central part of future internet research. In the IoT, heterogeneous devices are connected to sense the environment or to observe individual tasks. Many research fields use the seamless IoT infrastructure to interact with the integrated devices and diverse services. Furthermore, IoT is a promising technology to increase the comfort and quality of life and opens new ways of interaction between people and things. Real life applications in healthcare sectors, home automation, industry, smart cities, monitoring scenarios, etc. benefit from the low-cost wireless technology in IoT on one hand. On the contrary, IoT system has many challenging features: many devices are resource-constrained with energy and memory, are highly heterogeneous and their applications continuously transmit transient information. Furthermore, Requesting, delivering and updating the information in IoT are challenging because of the resource limitation. Named Data Networking (NDN) is one of the latest and the most important Information-Centric Networking (ICN) approaches which uses named data to deliver data in the network. Based on hierarchically structured names, NDN matches the application pattern of IoT systems and uses its communication concept to optimize the power supply and distribute the data efficiently in the network. This paper discusses the main concepts of NDN including naming, routing, forwarding and caching in IoT infrastructure. To achieve an efficient system in different applications scenario in future IoT, an IoT architecture is proposed via NDN called IoT-NDN. The objective of this research work is the design and development of IoT-NDN for different fields in IoT systems. The deployment of IoT-NDN is challenging and requires proper design choices. The proposed solution and challenges of all mentioned issues are discussed in this paper.

Index Terms—Internet of Things (IoT), Named Data Networking (NDN)

I. INTRODUCTION

Wireless communication technologies offer many advantages for applications and services which are deployed in various modern areas such as industry, healthcare, smart home, smart cities, etc. The high request for data from devices which are connected and integrated into the Internet and worldwide has inspired the Internet of Things (IoT). Currently, IoT systems can use the standard Internet protocol (IP) to connect the devices to the Internet. Also, many research groups have proposed protocols for the required wireless network of the IoT based on the IP. On the one hand, IP brings some advantages such as enabling the communication compatibility between devices in several domains. On the other hand,

deploying IP in the IoT and its devices is challenging [1] as introduced in more detail in III.

However, IoT systems have application and device requirements such as scalability, robustness, power efficiency. The available solutions which are based on the host-to-host IP paradigm influence the communication performance in current and future Internet. Using the IP means also using IP addresses for all devices connected to the Internet (need a big address space). That leads to the need for new protocols and algorithms to match the requirements of the IoT and its devices. For example, it is necessary to consider new resolution systems to translate application level requests such as the Uniform Resources Identifier (URI) into IP addresses. Furthermore, supporting new protocols for mobility in such systems is required. Additionally, the security of the end-to-end oriented devices is not entirely covered using the IP.

In the last years, the research community is exploring the communication paradigm of current Internet architecture. They suggest either clean-slate or cutting-edge approaches to replace current Internet architecture with a new architecture to achieve the requirements of today and modern communication paradigm. Named Data Networking (NDN) paradigm [2] has shown as a solution for this goal. In contrast to the IP, devices can specify what they are looking for without knowing the location of these data to be provided. NDN is a new communication model. Using NDN, the data are the first class of communication [2]. The data are a self-identifying unit which can be requested by sending an Interest packet (NDN message type) from any authorized consumer using location-independent and unique names. The NDN communication model has some features such as scalable naming scheme, lightweight configuration, management operations and simple communication model. Therefore, NDN has been suggested as a particularly promising solution for IoT systems [1]. IoT systems run many applications in different domains on heterogeneous and constrained devices that may request the same data regardless of their provenance. The challenges of the IoT ecosystems have motivated the author to develop a high-level IoT-NDN architecture for IoT domains using NDN approach. The developed system of IoT-NDN has a clear architecture which considers the IoT and NDN model which are presented in. IoT-NDN is considering the type of data exchange in the IoT, configuration and management layer of

application and services as well as different strategy layer which is responsible for transmission and forwarding of data. The designed and developed IoT-NDN system will enable the research community, industry, and different sectors to contribute to the future Internet of Things with the Named Data Networking paradigm.

The rest of the paper is organised into four sections as follows: Section II presents related works to the IoT and NDN approaches. Section III discusses the challenges of the IoT and analyzes the NDN approach for IoT systems. The architecture of the IoT-NDN is introduced in section IV. Finally, section V concludes the paper and outlooks the future work.

II. RELATED WORK

In the IoT vision, a billion devices are all the time connected to the Internet and are reachable anywhere using the IP. The Internet is connecting the physical world to collect data using IoT devices [3]. However, the scarcity of low-cost energy devices and the need for a robust addressing scheme for billion of devices make the development and deployment of IoT a very challenging task. Data-centric approaches can provide several benefits to the IoT world [4], [5], [6], [7].

Although NDN has been originally conceived to improve the communication structure of the Internet. Today, several research efforts are investigating the extension of NDN in wireless networks, like ad-hoc and wireless sensor networks [8], [3], [4], [5], [9]. In such scenarios, NDN can natively provide many benefits, ranging from mobility support and low-cost configuration to delay tolerance and opportunistic networking. At the same time, it has been argued that the integration of NDN in the IoT brings more efficiency, flexibility, and robustness of the system [1].

Moreover, native in-network caching approaches can improve the energy efficiency of the overall network. There are several IoT scenarios in which NDN can serve and also enable new applications; for example, one of the most promising application is the smart home domain. In [10], the initial design of a Building Management System (BMS) is sketched with a focus on naming and security aspects. The process of data collection from every sensor is not considered here, but it is assumed that data are already cached in the server. Additionally, IoT with NDN and cooperative caching has been discussed in many research works regarding the wireless IoT scenarios [11], [12].

Interest packet forwarding in NDN is discussed in many research works [5], [6], [13]. The authors investigate approaches for Named Data Networking (NDN) forwarding which can perform the network. Reaching large-scale flows for efficient forwarding in NDN was the primary goal of such research articles.

III. ANALYSIS OF IoT AND NDN

this section discusses the challenges of the IoT and analyzes the NDN approach for IoT systems. The challenges and the main issues facing the IoT system in current Internet architecture are addressed and analyzed in III-A. III-B investigates

the NDN as an alternative solution for mentioned issues of IoT system. Furthermore, this section analyzes the possibility to build IoT system based on the NDN approach. III-B5 discusses the routing scalability in the NDN and presents some approaches which propose a solution for the scalability using the NDN paradigm.

A. Challenges of the IoT

Using IoT system in private, public or business area is attached to technical challenges and issues which are listed as follows:

- 1) **The connectivity of IoT devices:** Connecting the IoT devices will be a big challenge for the future of the IoT using the structure of current communication paradigm. Currently, the communication is based on server-client or host-to-host paradigm to connect devices on the Internet. This communication model is sufficient for current IoT system with thousands of devices which are involved. The IoT is growing up, and the number of IoT devices is increasing to more than billions, and that leads to a bottleneck in the current host-to-host paradigm. Therefore, it is necessary to develop new communication paradigm which can handle such requirements and a significant amount of data delivered from the IoT devices. The efficient solution to this problem is to connect the IoT devices to a decentralizing network.
- 2) **Technology Standards:** The IoT technology standards including network protocols, communication protocols, and data-aggregation standards are the core of processing, handling and storing data which are collected from the IoT devices. But these standards also have challenges in term of the aggregation of data, the avoidance of sending the same data and the data caching mechanisms which increase the energy and memory demands of IoT system. Since the target of IoT is the availability of data from all IoT devices and all the time, current standards do not fit these requirements and are attached to increasing the overhead of the constrained devices. Considering caching algorithms in the current standard, the devices can not cache the data efficiently, and the consumer should send another message to request the same data which already requested before. This is not efficient and increases the number of transmitted packets in the network and also impairs the performance of the system. Furthermore, other protocols are needed to support the mobility of the IoT system more efficient. For example, if the connection on devices to the Internet is interrupted due to interference, the device should be able to collect only the missing packets after getting a connection again.
- 3) **Mobility:** The number of mobile devices, and the inter-connectivity and the reliance of consumers in a mobile environment are increasing and making the system very challenging. The mobility of IoT devices has also been investigated by many research works such as [14] in term of the design and the usage such devices in the

mobile environment. Furthermore, mobile devices are equipped with different technologies to capture a wide range of data which impact the energy consumption, connectivity, and sensing capability. [15] shows different aspects such as context, Energy availability, Internet access and connectivity which are changed with the mobility of the IoT devices: All mentioned aspects are challenging and should be considered in designing and implementing an efficient and mobile IoT system.

- 4) **Complexity and Integration Issues:** The integration and testing of IoT systems are challenging because IoT systems have large numbers of Application Programming Interfaces (APIs), numerous protocols and multiple platforms. Additionally, the rapid developing of the APIs does not consider the resource limitation of the IoT devices. The extensions and modifications of such systems require additional steps in term of development time and the chance of implementation errors. The IoT system architecture has to consider the resource requirements of IoT devices in term of integrating new technologies, protocols, hardware or software to the system.

B. NDN for the IoT

The following subsections discuss the open issues of NDN with the IoT such as NDN packet length, in-network caching, data aggregation protocol and naming problem.

1) **NDN Packet Length:** As mentioned in [2], the NDN packets are encoded in a TLV format and do not have a fixed packet length. Each field in the NDN Interest or Data packet is defined with arbitrary length.

The idea of variable size format for NDN packets allows flexibility for future protocols modifications such as adding new parameters or modify existing parameters. NDN has been developed for Internet communication such as video streaming, etc. and the memory space has not been considered during the design and development.

The constrained devices, e.g., in the IoT use transmission protocols which can only transmit less data (few KBytes) due to the space restriction of these devices. That means that the overhead of packet transmission in such environments will be increased due to increasing the number of transmitted packets in the network. Furthermore, transmitting more packets in such networks impacts the device energy. Since the allowed size of transmitted packets with resource constrained devices is small in comparison to the original NDN packets, a solution of this problem is presented in [16].

2) **Caching in IoT/NDN:** The IoT applications are continually producing data which are requested either by consumers in the same network or another network. This data sometimes need to be cached for a while or has to be updated to keep the information fresh. This task requires more cache size and energy on the IoT devices to store, send, receive and update the data. Furthermore, where and which data should be cached on the IoT devices are also important questions to avoid redundancy and saving energy on the constrained devices.

As mentioned in III-A, caching on the IoT devices is challenging because of their resource limitation such as memory, energy and computation power. For example, the RAM of such devices are limited and separated for all processes running on the devices such as the operating system, applications, and protocols.

A large part of the RAM may be reserved for the OS, the communication protocols, and applications which run on the devices. This means that only a small memory is remaining for caching the data. To run NDN (NDN code including the core implementation classes and functions, and application code) on IoT devices, there are memory challenges regarding the RAM and the remaining memory for the CS table.

However, the classification of memory capacity in IoT systems must be considered when designing a new IoT architecture based on the NDN communication protocol. In-network caching is a fundamental feature of NDN. The data can be available for consumers at every intermediate device and may reduce the number of transmitted messages in the network. Cache mechanisms bring additional features for IoT constrained devices such as sleep mode capabilities. For example, the Interest packet can be satisfied from devices which already cached the data and are near to the consumers, and the remaining devices can switch in the sleep mode. [11] presents experiments to show the relationship between in-network distributed caching strategies with NDN and sleep techniques to save the energy on IoT devices. The results in [11] show that a small amount of cache size (e.g., 5KB) can significantly increase the data availability in the network which brings advantages in term of the energy efficiency.

Furthermore, caching strategies and replacement policies are main parts of the caching schemes in NDN [2]. For example, replacement policies are responsible for keeping information about the CS due to make a place for new receiving Data packets if the CS is completely filled. Replacing or removing a Data packet from the CS based on the selected replacement policy is very important step using the NDN, especially on the constrained devices because of the small size of the CS. Such mechanisms should be developed and implemented on the devices which run NDN paradigm, and they also require additional memory.

3) **Data Aggregation in Wireless Networks:** For instance, a consumer is looking for data having the name `"/ndn/temperature"` and uses the exclude parameter to fetch all Data packets of the requested data. It is to assume that the temperature has three Data packets which should be sent to the consumer. After the consumer sends the first Interest packet, it receives the first Data packet from the network and sends a second Interest packet including the name of the first Data packet as exclude parameter. The Interest packet includes the excludes parameter specified, e.g., the name component `packet_1`. The data producer is now not allowed to return any Data packet with the prefix `"/ndn/temperature/packet_1"`.

This approach has disadvantages and increases the overhead significantly. An Interest packet must be sent to fetch one Data packet. Furthermore, iterative sending of Interest packets can

not fetch all Data packets from all devices in the network. If a device is located behind another device which has the required data in its CS, the Interest will not be forwarded. Additionally, the exclude parameter needs a significant space in the Interest packets which does not scale with a big number of devices in the network because of the limited length of packets. The packet length limitation is determined by the maximum transfer unit (MTU) of the physical layer. This concept does not match the constrained wireless network devices which are very limited with the MTU and energy.

Therefore, another request type is needed to satisfy such application requirements. IV-B1 explains a new concept including new protocol and services as solution for this problem.

4) **Naming Problems in Wireless Networks:** Today, the traditional services are location oriented. To support the idea of future Internet communication using hierarchical names, the traditional services must be changed to be name-centric services. It means that the services are using names to deliver requested data to the consumers. The NDN paradigm can support the placement and distribution of name-centric services in the wireless networks such as the IoT [17]. It facilitates the access to these services without knowing their location.

In addition, naming convention schemes are the key for the automatic network configuration [18] which are not covered in the name centric networking such as in NDN. For example, designing and developing a new device or sensor which should be integrated into an IoT environment based on NDN to deliver data, this device will need the network configuration, e.g., to assign names. In [18] presents the way to request the required network configuration and the concept to develop a name-centric services in wireless networks based on NDN.

The name semantic has not been specified in NDN, and the peregrination of valid names for limited access to devices is also not supported in NDN. This is comparable to the link-local and multi- or unicast addresses which need predefined addresses for automatic network configuration of wireless devices such as in the IoT.

A further problem using NDN naming scheme is that the used names are usually verbose and have not been designed for small and constrained wireless devices. Such names are designed for humans, so that are easily readable. Furthermore, these names cause to increase the overhead of the wireless channels of devices regarding the bandwidth, memory, and energy. [18] suggests an efficient solution of such problems.

5) **Routing Scalability in NDN:** NDN applications and services use directly the names to route the data. NDN routers keep track of the Interest packets until they are satisfied and all data which are forwarded or requested are cached in the CS. Routing scalability is essential for supporting a large number of devices on a large scale networks [19].

In NDN, the routing plane can be coordinated with the forwarding plane which can present a simple method to improve the routing scalability [19]. Now the question arises about the routing scalability using the concept of names in NDN. There are new approaches which discussed the routing

scalability in NDN such as [20]. These approaches propose a solution to address the NDN routing scalability concern.

IV. ARCHITECTURE OF IoT-NDN SYSTEM

The architecture of IoT-NDN is introduced in this section. This section also gives a brief description of every component in the IoT-NDN architecture. Fig. 1 illustrates all components of IoT-NDN system.

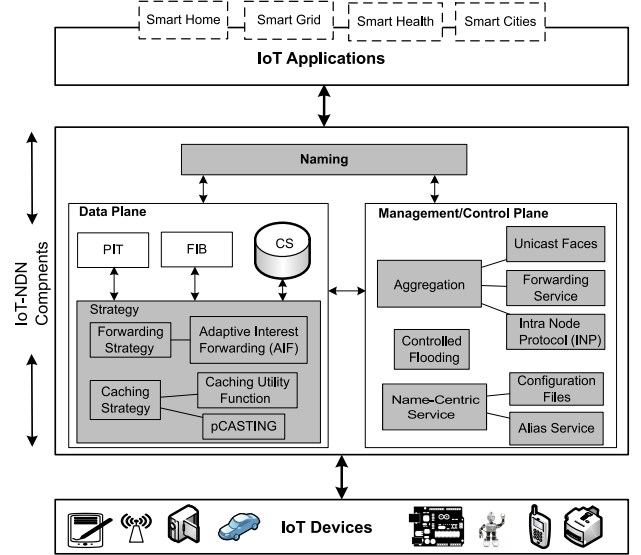


Fig. 1. IoT-NDN System architecture and its components

The IoT device layer is illustrated at the bottom of the IoT-NDN architecture which includes all heterogeneous IoT devices. The top of the IoT-NDN architecture contains the IoT applications which are requesting the data. IoT-NDN operates as a networking layer, and it serves to deliver the requested data from the underlying layer to the overlying layer.

The IoT-NDN has three main components: Naming, Data Plane and the Management and Control Plane. The naming component contains new naming schemes and structure which are suitable for the constrained wireless network devices. Management and control plane contains once again subcomponents such as Unicast Faces, Forwarding Services, Intra Node Protocol (INP), Controlled Flooding, Configuration and Alias Service. The data plan component contains the strategy which includes the caching and the forwarding strategies of IoT-NDN data. IoT-NDN components are responsible for handling the IoT-NDN packets and operate naming, caching, strategies, etc. in IoT-NDN system. The gray components which are illustrated in Fig. 1, are designed and developed to match the IoT requirements.

All devices which use IoT-NDN architecture also have three tables; CS, PIT, and FIB. All IoT-NDN packets (Interest and Data packets) are received and sent through faces. In IoT-NDN, the faces are used concerning the network or to local applications and simplify the processing and forwarding of messages in the IoT-NDN system. Face implementation for the IoT-NDN is explained in details in [2].

IoT-NDN architecture is the solution for the mentioned issues in section III. Tab. I summarizes all protocols and algorithms implemented and tested in the IoT-NDN. More details about the IoT-NDN components and their functionalities are explained in the following subsections.

TABLE I
IMPLEMENTED AND TESTED PROTOCOLS AND ALGORITHMS IN IoT-NDN

IoT-NDN Protocols & Algorithms	Implemented	Evaluated	References
Naming and NDN Protocols	✓	✓	[4], [18]
Name-Centric Services	✓	✓	[18]
Efficient Caching Algorithms	✓	✓	[6], [7]
Adaptive Interest Forwarding	✓	✓	[6]
Control Flooding	✓	✓	[6]
Data Aggregation Protocol	✓	✓	[5]
APIs for IoT-NDN	✓	✓	[21], [22]

A. Naming

The data in IoT-NDN are addressed by names. Requesting these data is based on hierarchically structured approach. The names contain human-friendly components and are location independent as shown in Fig. 2. Furthermore, IoT-NDN names describe a special task, an event or an application scenario. The architecture of IoT-NDN allows the applications to request specific data from producers or any device in the network using the naming mechanisms explained in this section.

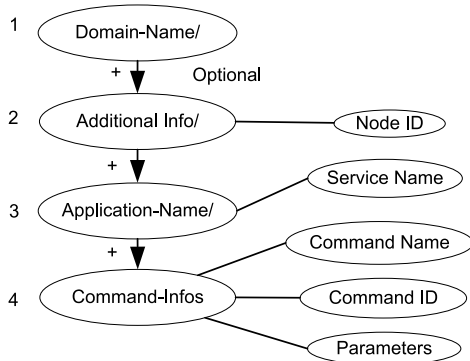


Fig. 2. Name Structure of the suggested Approach

Usually, the first component of the name is a global DOMAIN-Name which is assigned by the consumer. The second component is optional and could be used for sending additional information which could be interesting for some applications or services. Application and service name are assigned to the third component. The last component is for additional information such as device id.

The marker component in IoT-NDN names is used to deliver further information about an application, service or devices resources. IoT-NDN Names could flexibly represent additional information about applications or specific events by adding them to the last component of the name. For example, the timestamps could be added as a last component to the Interest packet and can be used as a versioning component for all

devices in the network. The device which receives the Interest packet can process the additional information from the last component. Furthermore, the marker component can be used as a command to request further and specific information. This feature of IoT-NDN name reduces the number of transmitted packets in the network and can be used in the routing layer to optimize the system. The evaluation results will be presented in [21].

B. Management and Control Plane

1) **Aggregation:** The aggregation, Forwarding Service, Unicast Faces and Intra Node Protocol (INP) components are explained in this section.

Data aggregation in wireless networks is very important because of the device limitation in term of the memory size of CS and energy. For example, forwarding fewer Interests and Data packets saves a large quantity of energy. Furthermore, data which have the same meaning or value could be aggregated in a Data packet and could be sent as an answer of the Interest packet to the consumer. More details about the aggregation and forwarding problems are explained in [5].

The concept of the data aggregation is to develop three components which are worked together in the IoT-NDN system. These components are Forwarding Service, Unicast Faces and the Intra Node Protocol (INP) which are presented in [5].

The radio service is extended to an additionally part called Unicast Face. To make the Unicast face working, the radio layer should be used device addressing scheme. It means that every radio packet should include the destination and source address in the local network. By receiving a broadcast packet from the neighbor device, the radio service generates a new Unicast service automatically. Furthermore, the radio service maps also the addresses between the new face and neighbor addresses. The generated Unicast Faces have a limited time dependency on the application. Every application can configure this time as needed. Additionally, the resources which reserved for the Unicast Faces on the device are released if the links are broken.

More information, implementation details and an example of the data aggregation approach are presented in [5].

2) **Controlled Flooding:** IoT-NDN devices are often battery-powered and sometimes mobile and communicate through the unreliable broadcast shared medium in a multi-hop fashion. In such environments, the Forwarding Information Base (FIB) tables are usually not populated in advance by routing information. The Interest packets are broadcasted hop-by-hop with controlled flooding mechanisms, as discussed in [13], [23] and reference therein. Flooding is indeed a simple and robust solution for finding a path to the requested data even under intermittent connectivity.

To reduce overhead and redundancy, the forwarding strategy is enhanced with timer-based packet suppression techniques that exploit packet overhearing, i.e., the device defers the packet of a specific delay and drops it if the same packet is overheard over the broadcast medium. The defer and listening time can be selected in different ways: it can be randomly

computed, or based on some specific information about the network topology, if available. For instance, if devices are aware of their distance to the producer, the forwarding scheme can prioritize the devices closer to the producer with shorter defer times [23].

Controlled flooding can also be used as the initial discovery phase to set a temporary FIB entry on the reverse path taken by the first Data packet [9]. After the producer is discovered, subsequent Interest packets can be Unicast using the established path, while flooding is used again only if the producer becomes unreachable. Unicast transmissions, however, hinder the possibility of sharing information over the wireless channel. In case more than one producer returns the data, the consumer can evaluate which path gives the best performance and send subsequent Interest only in that direction [24]. The mentioned mechanisms of the controlled flooding are designed for the IoT-NDN including the defer time functions for the Interest and Data packets.

3) **Name-Centric Service:** To let all IoT-NDN devices reachable from any device connected to the Internet, an IoT-NDN device runs as a gateway (see a brief description of the gateway in [18]) which has better performance in term of memory and energy. This gateway is developed to provide transition and services between the wired and wireless network. Furthermore, the gateway provides a protocol conversion from, e.g., the wired devices to the wireless devices such as in the IoT. Vice versa the application and service which are running on the devices using IoT-NDN should serve the requests and responses of other devices using this gateway.

All important configurations of name services and IoT-NDN devices including the gateway in the wireless network are also cached in the configuration component and provided by the gateway. Additionally, services on the gateway can be implemented as IoT-NDN applications which communicate with each other via the IoT-NDN daemon and face.

Since the IoT-NDN devices in the wireless network are constrained with memory and energy and not able to serve all name requests received from the Internet, the gateway runs a name alias service to solve this problem. The name alias service maps the verbose names obtained from the wired network (Internet) to the alias names used in the wireless network (e.g., the IoT) to fit the packet size on the constrained devices. The gateway has a list of all names with all integrated applications and services in the wireless network. By integrating a new service to the device or adding a new device to the network, an entry is added to a configuration file with the new service name. Furthermore, if any device in the network needs any information about the run services in the wireless network, then the gateway could satisfy this Interest packet. More details about the gateway implementation and structure which is used here is described in [25].

The alias service name algorithm is specially developed for the wireless network topology using the IoT-NDN system. In the joint work [18] from Torsten Teubler, Mohamed Hail, and Horst Hellbrueck, an algorithm is developed to map such names. There is a hash map storage or dictionary to map

verbose names to alias names. The authors in [18] develop a scheme to ensure that the aliases follow the order of the names. This scheme is consisting numbers as aliases and is assigned to the verbose names accordingly.

During the initialization of the IoT-NDN gateway, the name service requests the aliased service name and assigns the aliased name to the service. If the gateway receives an Interest packet from the Internet, the alias service maps this Interest packet to an alias number and forwards it to the wireless network.

C. Data Plane

1) **Strategy-In-Network Caching:** The Content Store (CS) is the caching place for the Data packets in IoT-NDN devices (comparable with the buffer with routers in today's Internet Protocol). Routers with IP protocol also buffer Data packets but can not reuse it again after forwarding them, while IoT-NDN devices are able to reuse the Data packets if they match the names of Interests. III-B5 discusses the routing scalability in the NDN. If an IoT-NDN device or router receives an Interest, it first looks in the CS for matching Data packets. The responsible algorithm of IoT-NDN checks the CS if a name of cached data is a prefix of the Interest name. If a match is found, then the Data packet is sent back to the consumer.

In IoT-NDN and other networks, different consumers are requesting the same data (multicast) which already requested before or could be lost during the transmission. Therefore, using the CS table can solve such problems. The dynamic data delivery can also benefit from caching the data in the CS.

Caching the Data packets in intermediate devices increases the cache hit ratio of data in the network.

The LRU is the default replacement policy in the NDN which is matching the concept of IoT-NDN caching algorithms. The LFU and random policies are additionally implemented in the IoT-NDN and are shown a good result and high performance nearly to LRU as well, see [7] and [26] for information.

Furthermore, a **probabilistic Caching Strategy (pCASTING)** is developed for the IoT-NDN devices which considers the data freshness and two device features, namely energy level, and storage capability. When a device receives a Data packet with a PIT matching, it performs the caching decision with the pCASTING strategy. That helps to adapt the caching probability of each device dynamically.

The way to compute the caching probability in pCASTING is very versatile. It means that the caching utility function can work with different routing protocols, and it can be easily modified by adding other parameters. Caching strategies and caching implementation are explained in more details in [7].

2) **Strategy-Forwarding:** The forwarding strategy component in IoT-NDN system takes the task to select the forwarding paths from the FIB to forward Interest packets. This component could also be used to control the traffic by remembering the number of unsatisfied Interests. Also, the forwarding component chooses the faces which through the

Interest packets could be forwarded using some performance metrics like delay or throughput. Such features are beneficial for the wireless network devices in term of select a specific forwarding or routing algorithm which matches better for the requirements of the constrained devices. IoT-NDN also supports the forwarding steps on devices and provides techniques for fast name Lookup.

Furthermore, forwarding strategy in the IoT-NDN supports the requesting of the lost Data packets in the network. It means if some of the Data packets are lost because of, e.g., the disconnection of the network communication, the IoT-NDN device can request the missed Data packets using some forwarding decision metrics. Using these metrics, the IoT-NDN devices or routers can also measure the performance of every face based on, e.g., the returning Data packets to select the suitable forwarding algorithms with low costs. Additionally, the IoT-NDN device can also use the InterestLifeTime parameter combined with another metrics of the Interest packet to request the missing Data packets. Implementation details of the forwarding decision metrics are explained in [6].

V. CONCLUSION

The first part of this paper is analyzing the challenges of the IoT and discussed issues in IoT system using the current communication paradigm based on IP. The analyzed NDN protocol in this paper is related to the cache mechanisms, data aggregation and naming in wireless networks. Notably, building an IoT system based on the NDN is challenging again because NDN has not been developed for constrained devices which are limited with energy and memory. After presenting the difficulty of using NDN for constrained devices in term of the integration of NDN into the IoT, the concept of the IoT-NDN system is introduced in IV which solves mentioned problems presented in this paper, summarized in Tab. I.

IoT-NDN also allows researchers to create an entirely new way of communication to access data using names in the future Internet and generates competitive advantages for industry and government sectors. The IoT-NDN system also provides modern mechanisms of communications regarding the transmission of data based on names.

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